

Exact Solution of Problem 10: Prime-Multiple Discrepancy

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Abstract

Problem 10 asks whether the discrepancy obtained by summing signs over the multiples of each prime is bounded. The answer is stronger than boundedness: for every nontrivial value of N , the exact optimum is 1. The proof below gives a detailed construction and verifies optimality.

1 Problem statement

For $N \geq 1$, write

$$[N] = \{1, 2, \dots, N\}.$$

Choose signs

$$\varepsilon_n \in \{-1, +1\}, \quad 1 \leq n \leq N.$$

For a prime $p \leq N$, consider the signed sum over all multiples of p in $[N]$:

$$\Delta_p(\varepsilon; N) = \sum_{\substack{1 \leq n \leq N \\ p|n}} \varepsilon_n.$$

Define

$$D(N) = \min_{\varepsilon_1, \dots, \varepsilon_N \in \{-1, +1\}} \max_{\substack{p \leq N \\ p \text{ prime}}} \left| \sum_{\substack{1 \leq n \leq N \\ p|n}} \varepsilon_n \right|.$$

The question is:

Is $D(N)$ bounded as $N \rightarrow \infty$?

The exact answer is as follows.

Theorem 1. *For every integer $N \geq 2$,*

$$D(N) = 1.$$

If one adopts the convention that the maximum over an empty set is 0, then $D(1) = 0$, since there are no primes $p \leq 1$.

2 The lower bound

We first show that discrepancy 0 is impossible once $N \geq 2$. This lower bound does not need any deep theorem about primes. The only facts used are that every integer greater than 1 has a prime divisor, and that a sum of an odd number of signs ± 1 cannot equal 0.

Lemma 1 (A prime with an odd number of multiples). *For every integer $N \geq 2$, there is a prime $p \leq N$ such that*

$$\left\lfloor \frac{N}{p} \right\rfloor$$

is odd.

Proof. We split into cases according to the parity of N .

First suppose $N = 2$. Then take $p = 2$. We have

$$\left\lfloor \frac{N}{p} \right\rfloor = \left\lfloor \frac{2}{2} \right\rfloor = 1,$$

which is odd.

Now suppose $N \geq 3$ is odd. Since $N > 1$, the integer N has at least one prime divisor: take the least divisor of N that is greater than 1; it must be prime. Choose such a prime and call it p . Because $p \mid N$, the quotient N/p is an integer. Also, since N is odd, every prime divisor of N is odd, and therefore the quotient N/p is odd as well. Hence

$$\left\lfloor \frac{N}{p} \right\rfloor = \frac{N}{p}$$

is odd.

Finally suppose $N \geq 4$ is even. Then $N - 1$ is an odd integer greater than 1. As above, $N - 1$ has a prime divisor; choose one and call it p . Then $p \leq N - 1 < N$, so $p \leq N$. Write

$$N - 1 = pt.$$

Because $N - 1$ is odd and p is odd, the quotient t is odd. Now

$$\frac{N}{p} = \frac{N - 1}{p} + \frac{1}{p} = t + \frac{1}{p}.$$

Since $p \geq 2$, we have $0 < 1/p < 1$. Therefore

$$\left\lfloor \frac{N}{p} \right\rfloor = \left\lfloor t + \frac{1}{p} \right\rfloor = t,$$

which is odd. This proves the lemma in all cases. □

Lemma 2. *For every $N \geq 2$, one has*

$$D(N) \geq 1.$$

Proof. Fix $N \geq 2$, and consider an arbitrary signing

$$\varepsilon_1, \dots, \varepsilon_N \in \{-1, +1\}.$$

By Lemma 1, choose a prime $p \leq N$ for which

$$k = \left\lfloor \frac{N}{p} \right\rfloor$$

is odd.

The multiples of p in $[N]$ are exactly

$$p, 2p, 3p, \dots, kp,$$

so the prime-multiple sum $\Delta_p(\varepsilon; N)$ contains exactly k signs. Since k is odd, this sum is a sum of an odd number of terms, each equal to $+1$ or -1 .

Such a sum cannot be 0. Indeed, if a of the k terms are equal to $+1$, then the remaining $k - a$ terms are equal to -1 , and the sum is

$$a - (k - a) = 2a - k.$$

The number $2a$ is even, while k is odd, so $2a - k$ is odd. In particular $2a - k \neq 0$. Therefore

$$\Delta_p(\varepsilon; N) \neq 0.$$

Since $\Delta_p(\varepsilon; N)$ is an integer, nonzero implies

$$|\Delta_p(\varepsilon; N)| \geq 1.$$

Thus, for this arbitrary signing,

$$\max_{\substack{r \leq N \\ r \text{ prime}}} |\Delta_r(\varepsilon; N)| \geq 1.$$

Because the signing was arbitrary, the minimum over all signings is also at least 1. Hence $D(N) \geq 1$. $\square$

3 The upper bound: an optimal signing

We now construct a signing whose discrepancy is at most 1. Since the lower bound already says that no signing can do better, this will prove Theorem 1.

The construction is the simple half-split signing:

$$\varepsilon_n = \begin{cases} +1, & 1 \leq n \leq N/2, \\ -1, & N/2 < n \leq N. \end{cases} \quad (3.1)$$

Equivalently, since n is an integer,

$$\varepsilon_n = \begin{cases} +1, & 1 \leq n \leq \lfloor N/2 \rfloor, \\ -1, & \lfloor N/2 \rfloor < n \leq N. \end{cases}$$

Thus the first half of $[N]$ receives sign $+1$, and the second half receives sign -1 .

Lemma 3. *Let $N \geq 1$ and let $p \leq N$ be a prime. Put*

$$k = \left\lfloor \frac{N}{p} \right\rfloor.$$

Then, among the multiples

$$p, 2p, 3p, \dots, kp$$

of p in $[N]$, exactly

$$\left\lfloor \frac{k}{2} \right\rfloor$$

lie in the first half $\{n \in [N] : n \leq N/2\}$, and exactly

$$\left\lceil \frac{k}{2} \right\rceil$$

lie in the second half $\{n \in [N] : n > N/2\}$.

Proof. The multiples of p in $[N]$ are precisely

$$p, 2p, \dots, kp,$$

because $kp \leq N$ and $(k+1)p > N$ by the definition of $k = \lfloor N/p \rfloor$.

A multiple mp lies in the first half if and only if

$$mp \leq \frac{N}{2}.$$

Equivalently,

$$m \leq \frac{N}{2p}.$$

Therefore the number of first-half multiples is

$$\left\lfloor \frac{N}{2p} \right\rfloor. \tag{3.2}$$

We now show that this equals $\lfloor k/2 \rfloor$.

Write the Euclidean division of N by p as

$$N = kp + r, \quad 0 \leq r < p.$$

Dividing by $2p$, we get

$$\frac{N}{2p} = \frac{k}{2} + \frac{r}{2p}. \tag{3.3}$$

Since $0 \leq r < p$, we have

$$0 \leq \frac{r}{2p} < \frac{1}{2}.$$

There are two cases.

First suppose $k = 2h$ is even. Then (3.3) gives

$$\frac{N}{2p} = h + \frac{r}{2p}, \quad 0 \leq \frac{r}{2p} < \frac{1}{2}.$$

Hence

$$h \leq \frac{N}{2p} < h + \frac{1}{2} < h + 1,$$

so

$$\left\lfloor \frac{N}{2p} \right\rfloor = h = \frac{k}{2} = \left\lfloor \frac{k}{2} \right\rfloor.$$

Next suppose $k = 2h + 1$ is odd. Then (3.3) gives

$$\frac{N}{2p} = h + \frac{1}{2} + \frac{r}{2p}, \quad 0 \leq \frac{r}{2p} < \frac{1}{2}.$$

Hence

$$h + \frac{1}{2} \leq \frac{N}{2p} < h + 1,$$

so

$$\left\lfloor \frac{N}{2p} \right\rfloor = h = \left\lfloor \frac{2h+1}{2} \right\rfloor = \left\lfloor \frac{k}{2} \right\rfloor.$$

Combining the even and odd cases, the number of first-half multiples is

$$\left\lfloor \frac{k}{2} \right\rfloor.$$

There are k multiples of p in total. Hence the number of second-half multiples is

$$k - \left\lfloor \frac{k}{2} \right\rfloor = \left\lceil \frac{k}{2} \right\rceil.$$

This proves the lemma. □

Lemma 4. *For every $N \geq 2$, one has*

$$D(N) \leq 1.$$

Proof. Use the half-split signing (3.1). Fix a prime $p \leq N$, and set

$$k = \left\lfloor \frac{N}{p} \right\rfloor.$$

By Lemma 3, among the multiples of p in $[N]$, the number with sign $+1$ is

$$\left\lfloor \frac{k}{2} \right\rfloor,$$

and the number with sign -1 is

$$\left\lceil \frac{k}{2} \right\rceil.$$

Therefore the signed sum over multiples of p is

$$\begin{aligned} \Delta_p(\varepsilon; N) &= \sum_{\substack{1 \leq n \leq N \\ p|n}} \varepsilon_n \\ &= \#\{\text{positive-signed multiples of } p\} - \#\{\text{negative-signed multiples of } p\} \\ &= \left\lfloor \frac{k}{2} \right\rfloor - \left\lceil \frac{k}{2} \right\rceil. \end{aligned}$$

Again there are two cases.

If $k = 2h$ is even, then

$$\left\lfloor \frac{k}{2} \right\rfloor = \left\lceil \frac{k}{2} \right\rceil = h,$$

so

$$\Delta_p(\varepsilon; N) = h - h = 0.$$

If $k = 2h + 1$ is odd, then

$$\left\lfloor \frac{k}{2} \right\rfloor = h, \quad \left\lceil \frac{k}{2} \right\rceil = h + 1,$$

so

$$\Delta_p(\varepsilon; N) = h - (h + 1) = -1.$$

Thus, for every prime $p \leq N$,

$$\Delta_p(\varepsilon; N) \in \{0, -1\}.$$

Consequently

$$|\Delta_p(\varepsilon; N)| \leq 1$$

for every prime $p \leq N$, and hence

$$\max_{\substack{p \leq N \\ p \text{ prime}}} |\Delta_p(\varepsilon; N)| \leq 1.$$

Since $D(N)$ is the minimum of this maximum over all possible signings, the existence of this particular signing proves

$$D(N) \leq 1.$$

□

4 Conclusion

By Lemma 2, every signing has prime-multiple discrepancy at least 1. By Lemma 4, the half-split signing has prime-multiple discrepancy at most 1. Therefore, for every $N \geq 2$,

$$1 \leq D(N) \leq 1,$$

and so

$$\boxed{D(N) = 1.}$$

This proves Theorem 1.

Remark 1. The previously recorded elementary solution gave $1 \leq D(N) \leq 2$. The half-split signing improves the upper bound from 2 to 1, which is best possible by the lower bound. The key point is that for each prime p , the multiples of p are split almost exactly in half by the cut at $N/2$: exactly $\lfloor k/2 \rfloor$ of the $k = \lfloor N/p \rfloor$ multiples lie on the positive side, and exactly $\lceil k/2 \rceil$ lie on the negative side.